

ABERDEEN 35 WNW, SD, USA

WMO: 726670

Lat: 45.7114N

Lon: 99.1296W

Elev: 1957

StdP: 13.69

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 54933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-11.8	-21.2	1.8	-15.3	-17.1	2.3	-11.4	29.8	15.2	27.0	14.2	7.5	N/A	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	89.0	69.7	85.5	69.0	82.4	67.2	74.2	83.2	72.0	81.5	69.8	79.2	9.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	124.8	79.3	68.9	114.1	77.6	66.6	105.2	75.1	39.1	83.2	37.0	81.8	35.1	79.2	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.3	22.1	19.2	DB	-24.3	94.6	5.3	4.0	-28.2	97.5	-31.3	99.8	-34.2	102.1	-38.1	104.9	(4)
			WB	-23.9	77.7	5.3	2.4	-27.7	79.4	-30.8	80.8	-33.8	82.2	-37.7	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	12.4	13.3	27.9	40.6	54.1	64.4	70.3	67.7	59.5	44.5	30.1	15.5	(6)
		DBStd	23.13	13.63	13.04	13.87	10.54	8.91	6.35	5.53	5.90	8.05	8.98	11.57	12.14	(7)
		HDD50	5162	1167	1029	695	310	61	1	0	14	218	598	1069		(8)
		HDD65	8871	1632	1449	1151	731	350	85	16	38	201	638	1046	1534	(9)
		CDD50	2183	0	0	9	29	189	432	628	548	298	47	3	0	(10)
		CDD65	417	0	0	0	0	14	66	180	121	35	1	0	0	(11)
		CDH74	4022	0	0	5	22	200	629	1647	1100	391	28	0	0	(12)
		CDH80	1233	0	0	0	3	37	154	558	365	111	5	0	0	(13)
(14)	Wind	WSAvg	8.7	8.9	8.9	9.9	11.1	9.8	7.7	6.8	6.9	8.2	9.2	8.8	8.6	(14)
(15)	Precipitation	PrecAvg	20.9	0.5	0.6	0.9	1.7	3.1	3.8	3.1	2.4	2.0	1.8	0.6	0.5	(15)
(16)		PrecMax	26.2	1.9	1.4	2.1	3.5	6.2	6.7	8.5	4.5	5.7	5.7	2.1	1.3	(16)
(17)		PrecMin	13.4	0.1	0.1	0.1	0.4	1.0	1.2	1.1	0.6	0.4	0.2	0.0	0.0	(17)
(18)		PrecStd	3.3	0.4	0.3	0.6	0.8	1.6	1.4	1.7	1.1	1.4	1.5	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	44.3	48.2	69.4	77.1	84.1	88.7	95.3	94.2	89.2	77.7	63.9	46.7		(19)
(20)		0.4% MCWB	36.7	40.8	51.9	54.9	62.7	68.1	70.5	69.9	67.0	58.8	50.4	39.5		(20)
(21)		2% DB	38.1	40.5	60.9	69.0	79.7	84.4	89.4	88.4	83.0	69.7	56.2	39.3		(21)
(22)		2% MCWB	33.6	35.2	48.3	51.4	62.1	67.7	70.8	69.6	66.0	54.1	45.5	34.0		(22)
(23)		5% DB	35.1	35.6	53.6	63.1	74.9	80.8	85.9	83.8	78.3	64.4	50.9	35.1		(23)
(24)		5% MCWB	32.0	32.3	44.3	48.7	59.4	65.4	70.3	68.0	63.0	51.7	42.8	31.8		(24)
(25)		10% DB	31.8	31.9	46.7	58.3	70.2	77.5	82.7	80.1	74.1	59.4	45.9	31.4		(25)
(26)		10% MCWB	29.9	29.7	40.5	46.1	56.9	64.3	69.0	66.2	61.0	49.9	39.7	29.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	38.2	43.7	56.4	58.3	68.2	73.5	78.5	75.4	71.4	60.2	51.1	40.0		(27)
(28)		0.4% MCDB	44.8	47.8	63.7	72.1	78.3	83.2	85.8	83.0	81.7	73.8	61.3	46.5		(28)
(29)		2% WB	34.3	36.3	50.2	53.3	64.5	70.7	75.6	72.6	68.4	56.8	46.8	35.3		(29)
(30)		2% MCDB	37.9	40.7	62.0	65.0	76.7	80.1	83.7	83.6	77.4	66.5	55.1	39.7		(30)
(31)		5% WB	32.3	33.1	45.2	49.9	61.9	68.3	73.4	70.4	66.0	52.9	43.1	32.3		(31)
(32)		5% MCDB	35.2	36.1	54.9	61.7	72.2	78.0	82.1	80.5	74.0	63.1	50.0	35.5		(32)
(33)		10% WB	30.2	29.9	40.9	47.2	58.9	66.2	71.0	68.4	63.4	50.0	39.7	29.8		(33)
(34)		10% MCDB	32.5	32.5	48.0	57.3	67.7	75.5	79.9	77.3	72.3	58.9	45.4	32.1		(34)
(35)	Mean Daily Temperature Range	MDBR	16.7	16.7	17.6	20.8	21.4	20.2	21.5	21.4	22.4	19.5	18.1	14.7		(35)
(36)		5% DB MCDBR	19.9	19.0	28.6	31.0	27.6	25.4	26.2	26.8	27.9	29.3	25.1	19.3		(36)
(37)		5% DB MCWBR	16.0	14.9	16.7	15.7	13.8	12.7	11.3	11.3	13.1	15.7	15.8	14.0		(37)
(38)		5% WB MCDBR	18.9	19.0	28.2	28.5	23.9	22.8	21.6	24.0	23.1	26.9	24.3	18.1		(38)
(39)		5% WB MCWBR	15.7	14.6	16.6	16.0	13.1	13.0	12.3	11.6	13.0	15.5	15.7	13.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.291	0.300	0.360	0.379	0.369	0.375	0.373	0.344	0.302	0.272	0.259		(40)
(41)		taud	2.388	2.371	2.447	2.287	2.316	2.395	2.397	2.382	2.443	2.528	2.503	2.449		(41)
(42)		Ebn at Noon	266	283	297	285	282	285	281	278	277	277	268	262		(42)
(43)		Edh at Noon	23	28	30	39	39	37	36	35	30	24	20	19		(43)
(44)	All-Sky Solar Radiation	RadAvg	559	902	1242	1584	1828	2052	2114	1766	1370	879	577	430		(44)
(45)		RadStd	49	94	103	117	148	131	89	148	85	108	66	28		(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page